

EDA Report

Introduction

Around the globe, people have applied for the Internship in Data Visualization opportunity at Excelerate, and we are among those who have accepted this role to enhance our data analytics skills by applying our knowledge to the respective datasets to bring up insights from the underlying dataset.

Project Title: User Insights

Project Deadline: 19th January 2024, 18:00 IST

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The datasets we have provided are following

UserData.csv, and OpportunityWise.csv

The main objective of our EDA is to find out user insights, such as what kind of opportunity users prefer to have and their demographic characteristics, which include the city, state, country, and also their educational background.

We believe that after analyzing the data, we can find the targeted area where the Excelerate team can put more effort into digital marketing to capture the region where they can grow and also bring the opportunity that people most sought.

Also, the number of requests for internships is accepted, rejected, and allocated in comparison to the application, and if data can speak about the chances of rejection, then we will find out too.

Data Overview

We have been provided two datasets: "User Data" and "Opportunity Data".

To gather insights from the data, we have used a combination of Python, Excel, and Power BI to leverage the best of them for the analysis.

So first and foremost, we came up with the numbers of rows and columns that are available in both datasets, and as per the analysis, we found the following shape of the data:

Shape of Opportunity Data: (rows: 20322, columns: 21)

Shape of User Data: (rows: 27562, columns: 8)

The Opportunity dataset contains 20,322 entries and 21 columns, including key information like Profile Id, Opportunity Id, Opportunity Name, and more.

Some columns, such as Gender, City, State, and Zip Code, have missing values.

Notably, columns related to Reward Amount, Badge Id, Badge Name, Skill Points Earned, and Skills Earned contain 2,521 non-null entries, indicating available information for those aspects.

The User dataset consists of 27,562 entries and 8 columns, featuring information such as Preferred Sponsors, Gender, Country, Degree, Sign-Up Date, City, Zip Code, and IsFrom Social Media.

Several columns, including Gender, Degree, City, Zip Code, and Country, contain missing values.

The 'Sign Up Date' column suggests a date format, but it is stored as an object.

Further data cleaning and exploration are necessary to handle missing values and convert data types for meaningful analysis in both datasets.

Column Analysis

At the time of the column analysis, we find the number of issues that bring up the biases to report.

There are numbers of missing values in both data especially if we look to the Opportunity Dataset columns like "Reward Amount", "Badge Id", "Badge Name", "Skill Points Earned" we find the null values are more compare to other columns.

In User Dataset Gender, Degree, City, and Zip are having approx. 10K of missing values, also while exploration we notice that users have typed the wrong information in Country, City, and Zip field like they put the India in country section, but they put some city of U.S, and zip/pin code of Nigeria, which makes that most challenging/untidy data.

Also, the zip codes don't match with the city, country.

Apart from that some of the users have put the wrong spelling or abbreviation of the city which create the multiple record of the same cities and countries

As of moment, we cannot see any relationship with both the dataset because of the missing common columns, but while pre-processing if we find some relationship by feature engineering, then we will sure come up with the any legit relationship.

If you see the uniqueness of Opportunity Dataset DataFrame contains diverse information.

Notably, there are 11,481 unique profiles, 33 distinct opportunities, and 3,141 cities. The data spans 108 countries, 5,189 zip codes,

Application dates vary across 20,277 records, and there are 968 different major entries.

The dataset encompasses 56 unique badge IDs and names, with varying reward amounts and skill points earned, contributing to a rich and varied dataset.

In User Datasets we can see unique 94 preferred sponsors, 4 gender categories, 169 unique countries, and 4-degree types. The dataset covers 27,561 sign-up dates, 4,727 cities, and 7,453 zip

codes. Additionally, it indicates whether sign-ups are from social media with 2 unique values. The dataset provides a comprehensive overview of user preferences and demographic.

Profile ID Analysis

As far as we dig the data, we cannot find the uniqueness of Profile IDs that we can combine in both the data, but when we go and check the Profile IDs in Opportunity Data, we see that the user has applied for the multiple for the opportunity, which is quite predictable.

We come up with a figure of 8,833 users who have applied for more than one opportunity, and 11,468 users who have just enrolled in single opportunities.

Also, we have not encountered any missing profile IDs, so we can more rely on the Profile IDs for further exploration to summarize the data.

The User Data PreferredSponsors column has no null value and consists of 94 unique values.

Opportunity Status Description

As per the business perspective, we are always interested to see the conversion from the traffic or users, and according to that, stakeholders can take further actions on the data.

Initially, we find out that the opportunity data has 20,301 rows of columns, and there are no null values present in the status description field.

After pivoting the data, we came up with the following result:

Applied	89 occurrences
Dropped Out	24 occurrences
Not Started	1324 occurrences
Rejected	726 occurrences
Rewards Award	2521 occurrences
Started	809 occurrences
Team Allocated	14188 occurrences
Withdraw	620 occurrences

Basic Statistics

In Opportunity datasets, most of the columns are categorical, but only a few of them, such as “Reward Amount” and “Skill Points Earned,” are in numeric format.

And when we applied the statistical method to it, we got the following results on the screen:.

	Reward Amount	Skill Points Earned
count	2521.000000	2521.000000
mean	1081.261404	1186.964697
std	927.251398	399.172150

	Reward Amount	Skill Points Earned
min	50.000000	10.000000
25%	500.000000	1182.000000
50%	500.000000	1182.000000
75%	2500.000000	1182.000000
max	2500.000000	1776.000000

Due to the unavailability of the numeric data in UserData we cannot proceed with the statistical method.

Initial Observation

When we started working on the data, first and foremost, our eyes caught the discrepancy in the Userdata where the city, country, and zip code were not present as required.

Sometimes the user has just filled in the dummy details, which was challenging for us to tackle, but we have manually removed them.

Also, we find out that when we give the user input to enter the details, there are high chances of inconsistency, like some will just put Delhi in the state and some will put New Delhi.

According to the aggregation in the Gender column, we are able to understand that males are highly applied for the program.

Additionally, the majority of Excelerate users are from India, then Nigeria, and Excelerate is well-liked by both undergraduates and graduates.

We noticed the maximum number of users signed up in the month of June was 7939, and slowly the numbers were dropping.

People are more into internship opportunities than any other opportunities.

Like this, we have seen most rewards being claimed in internship categories around \$17,65,000, with a reward of around \$2500.

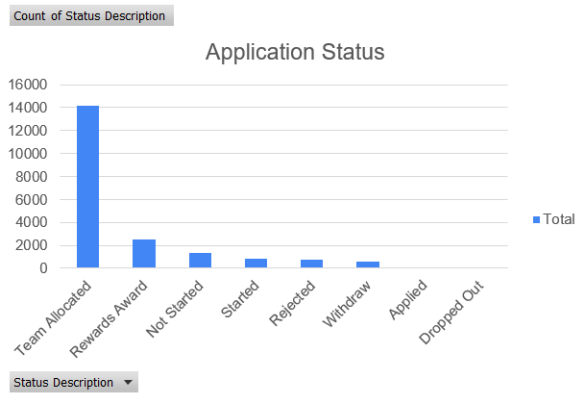
There are more things we can figure out from this data, which will be brought to the table in Week 2.

Visualizations

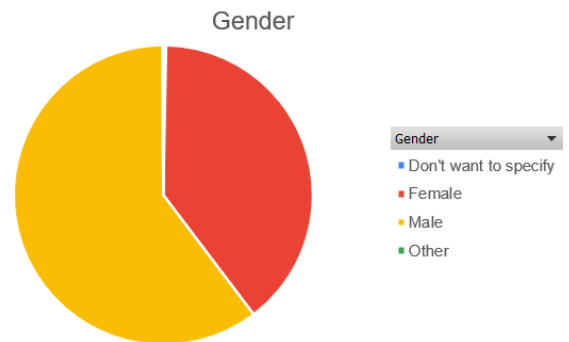
While in EDA, we have captured some of the visualizations to show the co-relation, a bar graph to show the number of users coming from social media, the status of the application, application segregation by gender, and many more.

Some of the visualizations are pasted here for understanding data insights.

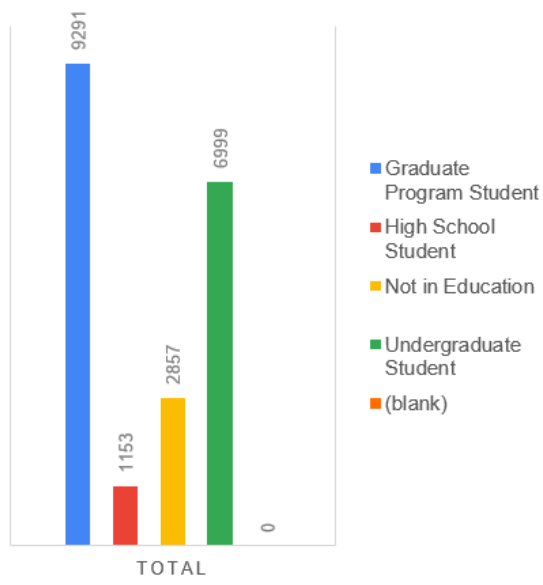
Status of Applications



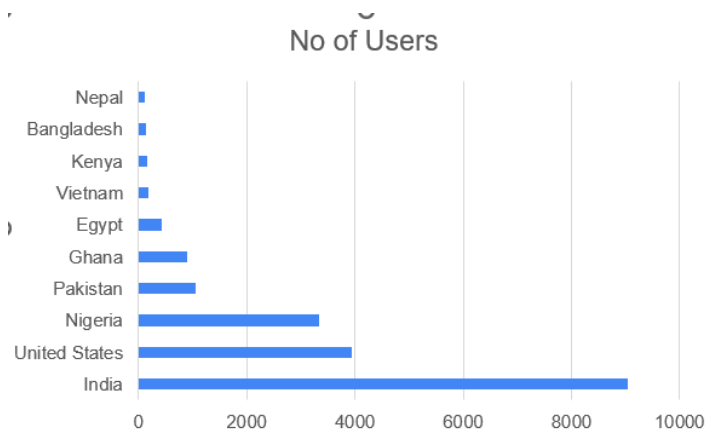
Gender



Intern Education



Top 10 Countries



Social Media

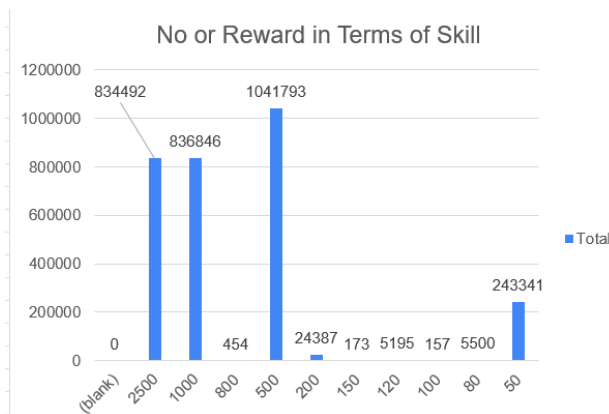
isFromSocialMedia

isFromSocialMedia
Boolean

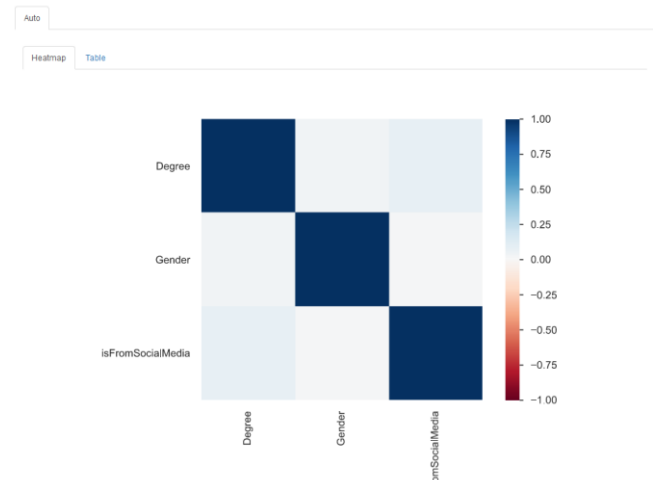
Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	27.0 KIB

True	13816
False	13746

Rewards in Terms of Skill



Co-relations



Challenged Faced

One major challenge was the lack of data quality/validation, as there were missing values and inconsistencies throughout the dataset.

Additionally, we faced multiple challenges while analyzing the data, especially since it's come to the point where we put our time into fixing the inconsistency in the city, country, and zip.

Where users have put the wrong data in the above-mentioned fields.

Apart from that, we have faced issues, but data was not enough to overcome them.

Next Step

In the next step, we will pre-process data to come up with more insights with the dashboard wireframe and show the insights with a neat and clean visualization.

This will involve cleaning and organizing the data, as well as transforming it into a format that is suitable for visualization.

Also, we will ensure that the dashboard wireframe is user-friendly and intuitive, allowing users to easily navigate and understand the insights presented.